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# A Multi-input Deep Neural Network Framework for Non-invasive Detection of Anemia using Finger Nail Images

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***Abstract*—Anemia, characterized by a deficiency in red blood corpuscles or hemoglobin, poses a significant global health challenge, particularly affecting vulnerable populations. Traditional diagnostic methods often involve invasive procedures, posing challenges in resource-limited settings. This study aims to explore non-invasive anemia detection using fingernail images and convolutional neural networks (CNNs) as a promising alternative to conventional diagnostic approaches. The study utilizes a dataset of fingernail images collected from hospitals in Ghana, comprising both anemic and non-anemic cases. The dataset undergoes preprocessing, including selective enhancement of red components, conversion to the CIELAB color space, and feature extraction. A multi-input Deep Neural Network (DNN) framework employing pre-trained CNNs is proposed for anemia classification. The pre-trained CNN architectures include EfficientNet B1, EfficientNet B4, and MobileNet V3. The framework's performance was assessed using two methodologies: The first involved random shuffling of the dataset, followed by division into training, testing, and validation sets, with evaluation metrics including Accuracy, Precision, F1 scores, and a Confusion Matrix. The second employed five-fold cross-validation, measured using accuracy. The evaluation of the proposed DNN framework using both of the methodologies indicates that EfficientNet B4 achieved the highest testing accuracy (97.87%), precision (97.88%), recall (97.87%), and F1 score (97.88%) and a cross-validation accuracy of 97.37% for the first and second methodologies respectively making it best fit for the proposed DNN framework. The findings demonstrate that the proposed framework yields promising results, especially under the second approach, and opens avenues for further exploration in transfer learning, fine-tuning of deep neural networks for multi-input feature integration, and cross-validation.**

***Index Terms*—Anemia, CNN, EfficientNet, MobileNet, Cross Validation.**

## I. Introduction

Anemia is a condition characterized by low red blood cell count or insufficient hemoglobin. Common causes include iron, vitamin deficiencies, chronic diseases, and genetic disorders like sickle cell anemia. Symptoms include paleskin, shortness of breath, fatigue and irregular heartbeat. Treatment varies based on the underlying cause and may involve dietary changes, supplements, or blood transfusions. Anemia can result from chronic infections, bone marrow disorders, and gastrointestinal bleeding. Timely diagnosis through blood tests is crucial for effective management. Addressing the root cause is essential to prevent complications. Pernicious anemia is linked to vitamin B12 deficiency. Inherited conditions like thalassemia and autoimmune disorders can also contribute to anemia. Early intervention and appropriate medical guidance are vital in managing and treating anemia effectively. This condition is especially common in young children in their early years and women undergoing menstruation, pregnancy, or postpartum recovery, accentuating its status as a severe public health concern.

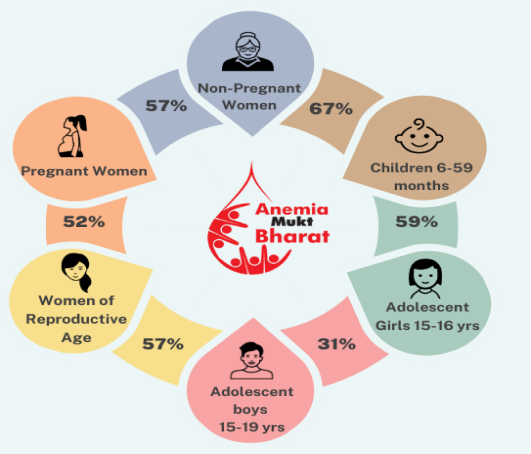



Fig. 1: Statistics of Anemia prevalence in different age groups of the people according to the statistics of NFHS conducted in 2019 [1].

Figure 1 demarcates the anemia prevalence in different age groups of the people in India conducted by the National Family Health Survey (NFHS) in 2019. The statistics clearly show that anemia is prevaling in different age groups of the people starting from children to aged persons. To curb this prevailance, Governement of India initiated a program called '*Anemia mukt Bharath*' in 2019. This program developed an $6 \times 6 \times 6$ strategy involving six groups of people by implementing six

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interventions in six institutions [1]. Traditionally, diagnosing anemia often involves invasive blood tests, posing risks such as infections and requiring substantial time, labor, and resources. To mitigate these issues, recent trends in medical research have turned towards non-invasive techniques, including machine learning algorithms and deep learning approaches [2]–[4], which hold promise for refining anemia diagnosis. These innovative approaches leverage the analysis of conjunctival pallor and other physical indicators that can be observed rapidly without blood draws. The advent of mobile technology and its application in health surveillance presents new frontiers in anemia screening. With their sophisticated imaging capabilities, smartphones represent a significant breakthrough in public health strategies by providing a convenient and easily accessible way to track possible anemia symptoms. A surge of research has been inspired by this shift toward technology-based solutions, with the goal of developing reliable, non-invasive techniques, utilizing computational analysis to identify and forecast anemia, thereby influencing public health interventions in the future.

In the current study, we propose a multi-input deep neural network (DNN) framework using deep learning that detects anemia from fingernail images. The top-performing model obtained an accuracy of 97% , proving to be a simple, non-invasive, cost-efficient approach for timely detection of anemia.

The rest of the paper is structured as follows: Section II provides the details of the existing literature on the classification task of anemia. The proposed model along with dataset details is discussed in Section III. Section IV discusses the results and discussions of the proposed model.

## II. Related Works

Several machine learning algorithms have been introduced for identifying anemia through their cost-effectiveness, user-friendly operation, and non-intrusive procedures. These methods rely on the visual assessment of paleness in tissues that become apparent when hemoglobin levels are low and red blood cell count is reduced. The tissues under observation for these signs are the palpebral conjunctiva, the bed of the fingernails, the palm, and the tongue.

In [5], the authors present a non-invasive method for estimating hemoglobin levels. This method analyzes color and texture from photographs of the patient's inner lower eyelid using a neural network that considers eight distinct features to make its assessment. The technique achieved a sensitivity and specificity of of 71.42% and 89.47% respectively, indicating its effectiveness in identifying anemia. A smartphone application called HemaApp, which noninvasively measures hemoglobin levels in blood using the phone's camera and various light sources, is presented in [6], which achieved high sensitivity (85.7%) and precision (76.5%), comparable to an FDA-approved non-invasive hemoglobin measurement device.

In [7], the utilization of facial images was explored for anemia detection, showcasing a novel approach in clinical diagnostics. The study leveraged deep learning techniques to analyze facial features, potentially offering a non-invasive method for anemia prediction in healthcare settings. A deep learning algorithm (DLA) to non-invasively monitor anemia using Electro-cardiograms is discussed in [8], which achieved an AUROC of 0.923 for 12-lead ECGs and 0.901 during internal and external validation. A comprehensive sensor-based system for monitoring the health of expecting mothers during pregnancy is proposed in [9] that detects non-invasive anemia, glucose rate, heart rate, fall detection, and body temperature monitoring, with Wi-Fi and GSM modules for telemonitoring. The model obtained the highest accuracy of 82.1% and AUROC of 0.824. Hybrid models, GA-SAE and GA-CNN, combined with genetic algorithms with Stacked Autoencoders (SAE) and Convolutional Neural Networks (CNN), are proposed in [10] to address nutritional anemia. The GA-CNN model excelled with an impressive accuracy of 98.50%. A mobile application for detecting anemia in children is presented by [11], where YOLOv5 is used. The system achieved a sensitivity and a specificity of 0.71 and 0.89 respectively. In [12], the authors have developed a device that employs a unique combination of LEDs and fiber optics in a multiwavelength spectrophotometry sensing platform with mechanical lever-operated sensor. The device, augmented with an algorithm, demonstrated an RMSE and a correlation of $1.47 \pm 0.042\, g\, dL^{-1}$ and 0.79 ± 0.03, respectively, between predicted and actual hemoglobin levels. In [13], the authors propose an affordable paper-based sensor coupled with a mobile application for on-the-spot identification of concentration of haemoglobin using a minimal blood sample. The device correlated highly with pathological gold standard results and excellent sensitivity and specificity for detecting healthy and anemic samples. In [14], an autonomous and cost-effective technique to monitor blood hemoglobin levels is proposed that quantifies hemoglobin levels based on color spectroscopy from the paleness of conjunctiva. Classification of anemia is done by comparing the estimated hemoglobin level with the threshold $<11.5\, g\, dL^{-1}$. The model achieved accuracy and sensitivity of $\pm 0.32\, g\, dL^{-1}$ and 89% compared to measured blood hemoglobin levels, respectively. Yet another non-invasive approach to anemia detection is presented in [15], which uses photographs of nails to detect leukonychia, an indicator of iron deficiency anemia, achieving an 89% accuracy in anemia detection with high sensitivity (96%). A similar approach to the non-invasive detection of anemia using AI techniques is discussed in [16], which achieved an accuracy of 97.8%. An affordable smartphone-based Hb analyzer (SHbA) is discussed in [17] that utilizes the phone's built-in ambient light detector to gauge the absorption of colorimetric Hb chemical tests within a compact setup, delivering results in under one minute demonstrating high sensitivity and specificity for anemia diagnosis. In [18] five deep neural network architectures UNet, UNet++, FCN, PSPNet, and LinkNet, are compared against their performance in segmentation of the regions of palpebral conjunctiva wherein LinkNet architecture outperformed its counterparts, achieving accuracy, intersection-over-union (IoU), and Dice score metrics of 94.17%, 90.14%,

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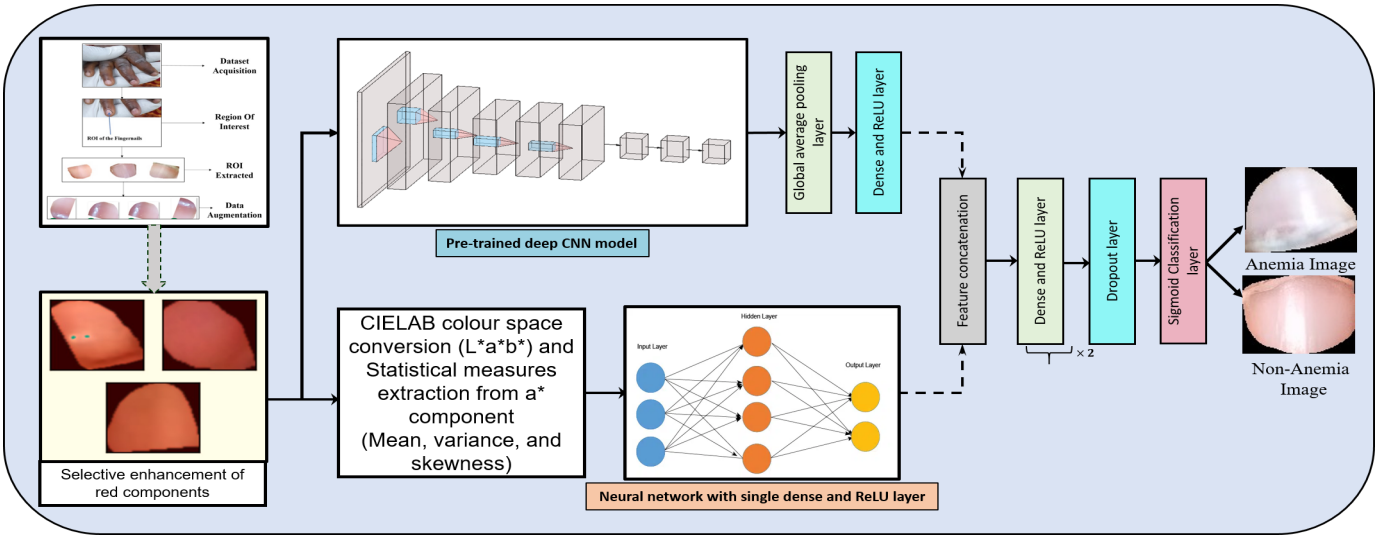



Fig. 2: Block diagram description of the proposed framework.

and 93.78%, respectively. In [19], the researchers propose a an artificial intelligence-enabled computational agent that can efficiently screen individuals for anemia, providing a cost-effective and accessible solution. The system employs segmentation and classification models, achieving an impressive Intersection Over Union (IOU) score of 0.922 for segmentation and high validation accuracy and recall for classification. In [20], a novel deep learning architecture was introduced that concurrently diagnoses anemia and determines hemoglobin concentration from images of the conjunctiva, offering a valuable resource for early anemia detection and treatment in resource-limited settings.

The current study proposes a multi-input deep neural network (DNN) framework that employs pre-trained convolutional neural network (CNN) models for detecting anemia from fingernail images. The framework was evaluated using two methodologies: (i) Random shuffling of the dataset, followed by division into training, testing, and validation sets, with evaluation metrics including Accuracy, Precision, F1 scores, and a Confusion Matrix. (ii) Five-fold cross-validation, measured using accuracy. The findings demonstrate that the proposed framework yields promising results, especially under the second approach, and opens avenues for further research in the current domain.

## III. Proposed Methodology

This section includes a description of the dataset, preprocessing of the images acquired, model architecture selection, proposed training, and evaluation framework. Figure 2 describes the workflow of the proposed multi-input DNN framework using pre-trained CNN models for anemia classification.

### A. *Dataset*

The dataset [21] used in this study comprises of fingernail images collected from several hospitals in Ghana and is publicly available under the Mendeley Dataset for detecting Anemia. The images were collected from young children aged five and below, and the region of interest was extracted using the threshold triangle approach. Initially, the dataset had 710 images, with 426 being anemic images and 284 non-anemic images. Later, the dataset was expanded using image data augmentation, which produced 4260 images. Of the total images produced, 2565 were anemic, and 1695 were non-anemic.

### B. *Image Preprocessing*

The ROI of images was enhanced selectively using Gaussian blur to enhance the red component while blurring the green and blue components. The enhanced images undergo conversion to (CIE $L^*a^*b^*$ ) color space model. The features pertaining to $a^*$ components: red components $a^*>0$ and green components $a^*<0$ are extracted using mean, standard deviation and skewness of the LAB images. These features along with enhanced images are given as input to the multi-input DNN framework for prediction. We are motivated by [22], [23] to utilize CIELAB color space in the proposed framework.

### C. *Model Architecture Selection and DNN Framework*

The preprocessed images were trained using the multi-input DNN framework. The framework receives two feature variables as input - the preprocessed images, and a vector containing features such as mean, standard deviation and skewness pertaining to a* component extracted from the preprocessed images for enhanced feature selection and extraction. The images are fed into an ImageNet pretrained CNN model, followed by a global average pooling layer and a dense ReLu layer for fine-tuning while the numerical a* feature vector is passed through a dense layer with ReLu activation function. The features obtained from both the channels are concatenated and passed through a sequence of two dense layers with Relu activation followed by dropout layer with dropout rate of 0.5 to reduce overfitting. Sigmoid activation was applied to the final layer for classification into the two classes namely Anaemic and Non-anaemic. The framework was individually tested using three different pretrained models. The pretrained

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CNN architectures used for the proposed framework include ImageNet pre-trained models: EfficientNet B1, EfficientNet B4 from the family of efficientNet models and third - MobileNet V3 , computationally efficient version of MobileNet architecture , a light weight model.

*1)* ***MobileNet V3***

The MobileNet family of models are known for their lightweight wight architecture ,less complex model size and computationally efficiency. MobileNet V3 was considered the current study. The model employs depthwise separable convolutions like its predecessors for reduced computational load and incorporates squeeze-and-excitation blocks, a lightweight attention mechanism. Despite its compact nature, the model maintains a competitive performance highlighting the model's effectiveness in anemia detection, making it a practical and resource-efficient solution for healthcare applications.

*2)* ***EfficientNet B1***

Two of the seven architectures of Efficientnet family of models were considered for the current study. The EfficientNet B1 architecture, known for its efficiency and accuracy, comprises multiple convolutional layers with depth-wise separable convolutions and feature scaling using compound scaling techniques. Additionally, it also uses MBConv blocks (Mobile Inverted Bottleneck Convolution), similar to MobileNetV2 but with squeeze-and-excitation optimization thus, offering a good balance between accuracy and efficiency.

*3)* ***EfficientNet B4***

The EfficientNet B4 architecture is renowned for its depth and complexity, featuring a stack of convolutional layers with depth-wise separable convolutions, multiple expansion and squeeze-and-excitation blocks, and an increased number of parameters compared to its predecessors. The model demonstrates exceptional performance with remarkable efficiency and testing accuracy, precision, recall, and F1 score and cross validation accuracy in anemia classification, despite its enhanced model complexity. This makes the model best suited for high-performance servers or devices where computational resources are not a primary constraint, and maximizing accuracy is key.

### *D.* ***Proposed training and Evaluation Framework***

The multi-input DNN framework was trained using two set of methodologies: (i) In the first approach the dataset was randomly shuffled and was split into three sets - training, validation and test sets. Model training was carried out by using train set and was tested on validation set for improving the performance in each epoch. Reduce learning rate on plateau and early stopping were used to monitor the performance of the model on the validation set in order to prevent overfitting. The weights pertaining best performing model in validation set was saved and was used further on test set for further predictions validating the model's efficiency. The performance of each of the three models were compared using metrics of evaluation such as accuracy, binary cross-entropy loss,precision, recall, F1-score and confusion matrix. (ii) The second approach employs k-fold cross validation on the dataset for obtaining an unbiased estimate of the model's performance. Five fold cross validation was performed for the current study. The entire dataset was randomly divided into five equal-sized subsets. Each of the three models was trained on four subsets with remaining subset as validation set for evaluating the models. This procedure was carried out for five times, with every iteration using a distinct subset for validation. Thus results from each fold were averaged to produce a single estimation. This approach provides a more robust way to assess model performance compared to a single train-test split, especially with limited data, It also helps in reducing bias in the estimation of model performance. The evaluation metrics primarily used for this include Cross validation accuracy and Binary cross-entropy Loss.

## IV. Experimental Setup and Results

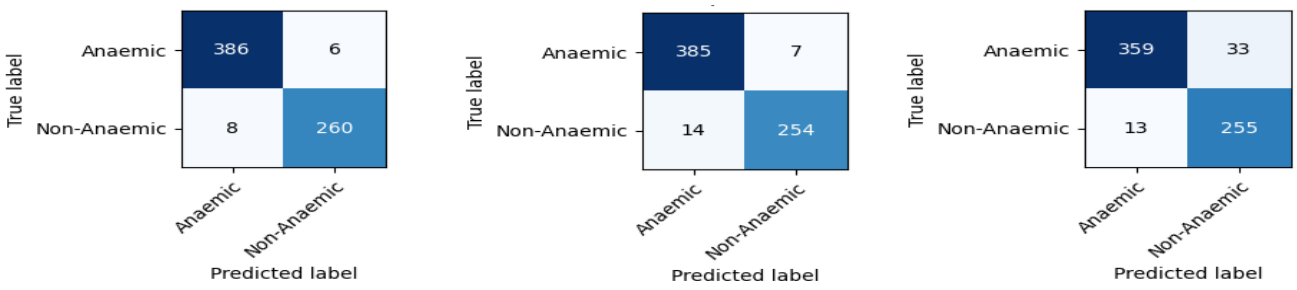
The proposed multi-input Deep Neural Network was trained in Kaggle environment with T4 GPU. The software requirements for implementing the multi-input Deep Neural Network framework utilizing the pretrained CNN architectures - EfficientNet B1, EfficientNet B4, and MobileNet are python 3.10, Keras and tensorflow for leveraging the pretrained CNN architectures and neural network fine-tuning, scikit learn library for leveraging different metrics of evaluation. The neural networks were trained for 30 epochs with a learning rate of 1e-4, batch size of 8. The multi-input Deep Neural Networks utilizing the pretrained CNN models - EfficientNet B4, EfficientNet B1 and MobileNet V3 would be referred to as DNN EffcientNet B4, DNN EfficientNet B1 and DNN MobileNet V3 for brevity. The results of the proposed framework using the two different methods of training and evaluation are as discussed below.

TABLE I: Results of the proposed DNN framework using first methodology against Evaluation Metrics

| Model | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| MobileNet V3 | 93.03 | 93.27 | 93.03 | 93.06 |
| **EfficientNet B4** | **97.88** | **97.88** | **97.87** | **97.88** |
| EfficientNet B1 | 96.82 | 96.83 | 96.82 | 96.81 |

Table I illustrates the outcomes of the proposed framework employing the first approach, benchmarked against four standard metrics: Accuracy, Recall, Precision, and F1-score. Meanwhile, Figure 3. depicts the corresponding confusion matrices obtained for the three models, providing a visual representation of their performance. Accuracy measures the proportion of correctly classified anemic and non-anemic cases among all instances. Precision indicates the proportion of correctly predicted anemic cases out of all instances predicted as anemic. Recall measures the proportion of actual anemic cases that were correctly identified by the model. The F1 score represents the harmonic mean of precision and recall, providing a balanced measure of model performance. The results clearly illustrate the impact of architectural variations on performance. The DNN EfficientNet B4, characterized by its model design and complexity, exhibited the highest testing accuracy of 97.87%, alongside precision, recall, and F1 score values of 97.88%, 97.87%, and 97.88%, respectively. The confusion matrix further validates this by providing a measure

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(a) (b) (c)

Fig. 3: Confusion charts of proposed DNN using (a) EfficientNet B4 (b) EfficientNet B1 (c) MobileNet V3.

of True positive, True Negative, False positive and False Negative. EfficientNet B4 3a shows highest True positive score and lowest true negative scores. The exceptional performance of EfficientNet B4 can be attributed to its extensive model capacity, which allows it to capture intricate patterns in the input data. On the other hand, The DNN EfficientNet B1, characterized by the architecture's balance between computational efficiency and model complexity achieved a testing accuracy of 96.82%, demonstrating competitive precision, recall, and F1 score values. The confusion matrix 3b gives more detailed insights that suggest the DNN performs well but with slightly more False Negative than DNN EfficientNet B4 making it bit more conservative in predicting anemia, preferring to minimize False Positive at the risk of increasing FN. In contrast,the deep neural network utilizing MobileNet V3, known for its lightweight design, achieved a testing accuracy of 93.03% with precision, recall, and F1 score values of 93.27%, 93.03%, and 93.06%, respectively. The confusion matrix 3c shows that the model despite achieving comparative True positive and true negative values, has the highest number of False postives indicating a tendency to over-diagnose anemia.

TABLE II: Results of the proposed DNN framework using second methodology against Evaluation Metrics

| Model | Metrics | Fold 1 | Fold 2 | Fold 3 | Fold 4 | Fold 5 | Effective Score |
|---|---|---|---|---|---|---|---|
| **EfficientNet B4** | **Accuracy** | **97.88** | **98.01** | **96.36** | **96.59** | **98.01** | **97.37** |
| | **Loss** | **0.08** | **0.04** | **0.16** | **0.12** | **0.14** | **0.11** |
| EfficientNet B1 | Accuracy | 95.3 | 96.12 | 98.12 | 97.88 | 97.06 | 96.9 |
| | Loss | 0.23 | 0.17 | 0.04 | 0.06 | 0.11 | 0.12 |
| MobileNet V3 | Accuracy | 96.47 | 96.94 | 97.41 | 95.18 | 95.3 | 96.26 |
| | Loss | 0.28 | 0.19 | 0.2 | 0.16 | 0.12 | 0.2 |

Table II, illustrates the results of the proposed framework using the second approach against the metrics: cross validation accuracy and loss. The proposed DNN EfficientNet B4, again surpasses the other models maintaining high consistency across all the folds with an effective cross validation accuracy of 97.37 and loss of 0.11. The DNN Efficient B1 and MobileNet V3 architecture maintain competitive performance with the former achieving effective cross validation accuracy of 96.90%with fold scores ranging between 95.30% and 98.12% and the latter with scores ranging from 95.18% to 97.41%, averaging at 96.26%. This performance, while strong, is slightly lower than that of EfficientNet B4. The effective loss score of DNN EfficientNet B1 indicates a slightly less optimal fit to the dataset due to slightly higher loss values compared to EfficientNet B4. However, the values are still relatively low, showing good model performance. The DNN MobileNet V3, despite having slightly higher loss values, still maintains good accuracy. However, higher loss values compared to both EfficientNet models, suggests that the model may not fit the data as closely as the EfficientNet models inspite of maintaining consistent accuracy.

TABLE III: Comparison of existing studies with the proposed framework

| **Methodology** | **Results** |
|---|---|
| EfficientNet models [24] | Accuracy: 97.52% |
| Analysis of patient videos using Deep learning [25] | Sensitivity: 92.52% |
| Detection of Anaemia from palpebral conjuctiva using CNN [26] | Accuracy: 94% |
| Proposed DNN Framework | Accuracy: 97.87% |

The findings clearly establish EfficientNet B4 as the most accurate model for the proposed DNN framework in the current study owing to its exceptional performance using both the approaches. The exceptional performance of family of EfficientNet models is also validated in [24]. Furthermore, consistent and an enhanced performance of the DNNs utilizing the pre-trained models: EfficientNet B1 and MobileNet V3 models using the second methodology indicate the efficiency of cross validation in assessing the performance of machine learning models paving way for further research and exploration in this field. Table III illustrates comparison of the results of the proposed DNN framework with the existing studies. The exceptional performance of the proposed multi-input DNN framework can be attributed to the transfer learning approach, coupled with enhanced feature extraction using multi-input channel and fine-tuning, thus emphasizing on the efficiency and reliability of the system for anemia detection and other health care applications.They also reiterate the effectiveness of machine learning approaches in non-invasive diagnosis.

## V. Conclusion

This study explored the potential of machine learning and medical image analysis to address anemia detection, offering a non-invasive and cost-effective alternative to traditional

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blood-based diagnostic methods. A multi-input deep neural network framework empowered by pre-trained convolutional neural network models was proposed for classifying anemia based on fingernail images. The framework was trained and evaluated using two methodologies: (i) random shuffling and evaluation on test set using accuracy, precision, recall and confusion matrix (ii) five fold cross validation and evaluation using accuracy and loss. The proposed DNN framework utilizing EfficientNet B4 exhibited exceptional performance using both the methodologies, underscoring its suitability for high-precision applications, albeit with increased computational requirements. The DNN EfficientNet B1 provided a balanced trade-off between accuracy and efficiency, while MobileNet V3 proved adept at resource-constrained scenarios. These findings contribute to the ongoing efforts to combat anemia by harnessing the power of machine learning and medical imaging. Incorporation of quality assessment [27]–[29] and preprocessing modules along with deep CNNs are important future tasks to explore.

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